PHILO (JIAQI) WANG

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PROFILE

Ph.D. researcher and applied AI developer specializing in LLM agents, human-AI interaction, behavioral evaluation, and full-stack AI systems. Experience across research design, system development, human-subject studies, data analysis, and academic writing.

EDUCATION

University of Illinois Urbana-Champaign — Ph.D. Information Sciences, Expected May 2029
University of Illinois Urbana-Champaign — M.S. Library & Information Science, 2024
Beijing Forestry University — B.A. Electronic Commerce, 2022

TECHNICAL SKILLS

AI & LLM: LLM agents, multi-agent systems, RAG, PyTorch, Hugging Face Transformers, vLLM, AutoGen, LoRA, supervised fine-tuning, prompt engineering
Programming & Data: Python, SQL, Java, JavaScript, C#, R, Pandas, NumPy, statistical modeling, A/B testing
Systems & Tools: FastAPI, Next.js, Node.js, PostgreSQL, MySQL, Firebase, SLURM, A100 GPUs, Git/GitHub, Unity
Research: Experimental design, interviews, think-aloud studies, qualitative coding, behavioral analytics, IRB, Qualtrics, REDCap

RESEARCH EXPERIENCE

Research Assistant — ActionCoach (NMSS-Funded Project) Aug. 2025 – Present
Python, AutoGen, FastAPI, Next.js, OpenAI API, Databases

- Built an LLM-based behavioral-activation training system with specialized agents for dialogue management, guidance, and safety.
- Developed full-stack workflows spanning prompt engineering, conversation-state management, database integration, and participant-facing interfaces.
- Coordinated expert interview and think-aloud study logistics; supported IRB modifications, REDCap identifiers, and participant workflows.

AI Agents in Collaborative Learning 2026 – Present
LLM Agents, OpenAI API, Behavioral Analysis, Learning Sciences

- Designed LLM peer agents with minimal-social, reactive, and proactive behaviors for an online collaborative learning environment.
- Developed interventions for inactivity, participation imbalance, and premature agreement in two-student collaborative learning tasks.
- Created an initiative–uptake coding framework and explored LLM-based automation of interaction coding.

LLM-as-a-Judge for Behavioral Annotation 2026
Python, Qwen3-14B, Hugging Face, Unsloth, LoRA, RAG, SLURM, A100/H100 GPUs

- Fine-tuned Qwen3-14B and compared supervised fine-tuning, RAG, and prompting as LLM-as-a-judge methods for discourse annotation on SLURM-managed A100/H100 GPUs.

SELECTED PROJECTS

Unveil / AI Playtesters — LLM Agent Behavioral Simulation
Python, vLLM, Hugging Face, Vector Retrieval

- Developed an evaluation framework comparing LLM-agent trajectories with behavioral data from approximately 95 human players.

· Evaluated multiple models across reasoning and memory configurations using escape rate, attention allocation, exploration breadth, action sequences, revisitation, and notebook use.

· Analyzed long-horizon failure modes across model, reasoning, and memory configurations.

Unveil — Misinformation Educational Escape Room
JavaScript, Node.js, MySQL, Python, R

· Developed and studied a narrative puzzle game teaching lateral reading, source evaluation, corroboration, and contextualization.

· Analyzed 10,000+ behavioral logs to examine player strategies, information behavior, and learning outcomes.

CEASAR — AR vs. Tablet Collaborative Learning
Data Analysis, Data Visualization

· Analyzed and visualized collaboration data from augmented-reality and tablet learning conditions to support comparisons of learner interaction patterns.

VR Navigation and Exploration Game
Virtual Reality, Eye Tracking, Navigation Analytics

· Helped design a task-based virtual environment for studying learning, decision-making, navigation, and information behavior through eye-tracking and movement data.

TEACHING & EDUCATIONAL PROGRAMS

Graduate Teaching Assistant — IS 596 and IS 504
University of Illinois Urbana-Champaign

· Supported class discussions, course activities, student questions, learning, and mentoring.

Summer Instructor — Fab Lab Youth Programs Summers 2025–2026
University of Illinois Urbana-Champaign

· Facilitated hands-on technology, making, coding, and digital-learning activities for students ages 9–13.

HUMAN-SUBJECTS RESEARCH

IRB submissions and modifications, participant recruitment and consent, Qualtrics, REDCap, interviews, think-aloud studies, Zoom experiments, qualitative coding, behavioral-log analysis, statistical analysis, data management, and academic writing.

SELECTED PUBLICATIONS

- **Wang, P.**, Tomaszewski, T., et al. "AI Playtesters: Evaluating LLM Agents as Diagnostic Instruments for Game Design." *Manuscript in revision.*
- Zhou, Y., **Wang, P.**, & Kang, J. (2024). "Navigating the Sky Together: Investigating Collaboration Dynamics through Annotation in an Immersive Learning Environment." *EDM 2024.*
- Liu, C., Ai, M., **Wang, P.**, & Adler, R. F. (2026). "No Sound, No Problem: Accessibility in Gaming for Deaf and Hard of Hearing Players." *PACM HCI / CHI PLAY 2026.*